



Security

Documentation and resources for helping organizations secure their compute environments, protect data, and comply with regulations using Google Cloud products.

Read documentation (<https://docs.cloud.google.com/docs/security/overview/whitepaper>)

Start your proof of concept with \$300 in free credit

- Develop with our latest Generative AI models and tools.
- Get free usage of 20+ popular products, including Compute Engine and AI APIs.
- No automatic charges, no commitment.

Get started for free (<https://console.cloud.google.com/freetrial>)

Explore Security in Google Cloud

Read documentation and Cloud Architecture Center articles about security products and procedures.



(/docs/security/overview/whitepaper)

[Google Cloud security overview](#)

(/docs/security/overview/whitepaper)



(/iam/docs/overview)

[IAM overview](#)

(/iam/docs/overview)

Learn how IAM works in Google Cloud and how



(/organization-policy/overview#how_organization_policy_works)

[How organization](#)

Learn the physical, administrative, and technical controls we use to help protect your organization's data.

you can use it to manage access.

[policy works](#)
(/organization-policy/overview#how_organization_policy_works)

Learn what organization policies and constraints are.

[More](#) ▾

Training, blog articles, and more

Go to training courses, blog articles, and other related resources.



(<https://www.cloudskillsboost.google/paths/15>)

[Security engineer learning path](#)

(<https://www.cloudskillsboost.google/paths/15>)

Study how to develop, implement, and monitor your organization's security infrastructure.



(<https://www.cloudskillsboost.google/paths/15>)



(<https://cloud.withgoogle.com/cloudsecurity/podcast/>)

[Cloud Security Podcast](#)

(<https://cloud.withgoogle.com/cloudsecurity/podcast/>)

Listen to industry experts talk about some of the most interesting areas of cloud security.



(<https://cloud.withgoogle.com/cloudsecurity/podcast/>)



(<https://cloud.google.com/blog/products/identity-security>)

[Cloud Security Blog](#)

(<https://cloud.google.com/blog/products/identity-security>)

Read the latest blog posts about Google Cloud security benefits and customer stories.



(/solutions/security)

[More](#) 

Security products by use case

Expand sections or use the filter to find products and guides for typical use cases.

Filter here

Security operations

Detect vulnerabilities, threats, and misconfigurations.



[Advisory Notifications](/advisory-notifications/docs) (/advisory-notifications/docs)

Receive well-targeted, timely, and compliant communications about security and privacy events in the Google Cloud console.



[Cyber Insurance Hub](/risk-manager/docs) (/risk-manager/docs)

Evaluate your organization's security posture and connect with insurance partners to obtain exclusive cyber insurance coverage and personalized pricing.



[Google Security Operations](/chronicle/docs) (/chronicle/docs)

Detect, investigate, and respond to cyber threats with SIEM and SOAR technology. Extract signals to find threats and automate the response.



[Google Threat Intelligence](https://gtidocs.virustotal.com) (https://gtidocs.virustotal.com)

Know who's targeting your organization with unparalleled visibility into the global threat landscape.

[Model Armor](/model-armor) (/model-armor)

Enhance AI security and safety by screening LLM prompts and responses.



[Security Command Center](/security-command-center/docs) (/security-command-center/docs)

Understand your security and data attack surface.

Access management



Provide unified, federated identity with least privilege policies to reduce the risk of data breaches and other security incidents.



[Access Context Manager](/access-context-manager/docs) (/access-context-manager/docs)

Allow organization administrators to define fine-grained, attribute-based access control for projects and resources in Google Cloud.



[Certificate Authority Service](/certificate-authority-service/docs) (/certificate-authority-service/docs)

Simplify, automate, and customize the deployment, management, and security of private certificate authorities (CA).



Identity and Access Management (IAM) (/iam/docs)

Establish fine-grained identity and access management for Google Cloud resources.



Identity and Access Management (IAM) Recommender

(/policy-intelligence/docs/role-recommendations-overview)

Identify excess permissions using policy insights.



Plan identity and access management (/architecture/identity)

Plan your design for granting the right individuals access to the right resources for the right reasons. [🔗 \(/architecture/identity\)](/architecture/identity)



Ensure access and identity

(https://www.cloudskillsboost.google/course_templates/645)

Study fundamental features of cloud security related to access management and identity.

[🔗 \(https://www.cloudskillsboost.google/course_templates/645\)](https://www.cloudskillsboost.google/course_templates/645)



Security and IAM planning resources (/architecture/security-iam)

Plan your approach with Architecture Center resources across a variety of identity and access management topics. [🔗 \(/architecture/security-iam\)](/architecture/security-iam)

Application security



Protect your workloads against denial-of-service attacks, web application attacks, and other security threats.



[Binary Authorization](/binary-authorization/docs) (/binary-authorization/docs)

Deploy only trusted containers on Google Kubernetes Engine.



[Certificate Manager](/certificate-manager/docs) (/certificate-manager/docs)

Acquire and manage TLS (SSL) certificates for use with Cloud Load Balancing and Media CDN.



[Identity-Aware Proxy \(IAP\)](/iap/docs) (/iap/docs)

Manage access to applications running in App Engine standard environment, App Engine flexible environment, Compute Engine, and GKE.



[Fraud Defense](/recaptcha-enterprise/docs) (/recaptcha-enterprise/docs)

Protect your organization's website from fraudulent activity, spam, and abuse.



[Secure Web Proxy](/secure-web-proxy/docs) (/secure-web-proxy/docs)

Migrate to Google Cloud while keeping your organization's existing security policies and requirements for outbound web traffic.



Web Risk (/web-risk/docs)

Detect malicious URLs on your organization's website and in client applications.



Google Cloud Armor (/armor/docs)

Help protect your services against DoS and web attacks.



Cloud Load Balancing (/load-balancing/docs)

Scale and distribute app access with high-performance load balancing.



Detect malicious URLs in Web Risk (/web-risk/docs/detect-malicious-urls)

Follow guidance to install and run the sample app to detect malicious URLs in a Go environment.

Auditing, monitoring, and logging



Collect, store, analyze, and monitor your organization's aggregated platform and system logs with a comprehensive solution.



Access Transparency

(/cloud-provider-access-management/access-transparency/docs)

Get visibility over your organization's cloud provider through near real-time logs.



[Audit Manager](/audit-manager/docs) (/audit-manager/docs)

Provides assessments about your adherence to compliance requirements.



[Cloud Audit Logs](/logging/docs/audit) (/logging/docs/audit)

Gain visibility into who did what, when, and where for all user activity on Google Cloud.



[Cloud provider access management](/assured-workloads/cloud-provider-access-management/docs)

(/assured-workloads/cloud-provider-access-management/docs)

Use this group of products for progressively greater transparency and control over access to your content stored in Google Cloud.



[Endpoint Verification](/endpoint-verification/docs) (/endpoint-verification/docs)

Create an inventory of devices running Chrome OS and Chrome Browser that access your organization's data.



[Personalized Service Health](/service-health/docs) (/service-health/docs)

Identify Google Cloud service disruptions relevant to your projects so you can manage and respond to them efficiently.



Unified Maintenance (/unified-maintenance/docs)

Manage planned maintenance across Google Cloud services.



Cloud Logging (/logging/docs)

Store, search, analyze, monitor, and alert on log data and events from Google Cloud and AWS.



Cloud Monitoring (/monitoring/docs)

Get visibility into the performance, availability, and overall health of cloud-powered applications.



Network Intelligence Center (/network-intelligence-center/docs)

Use a single console for comprehensive network monitoring, verification, and optimization.

Cloud governance



Manage your resources in a secure and compliant way with visibility and control over your cloud environment.



Assured Workloads (/assured-workloads/docs)

Secure your workloads and accelerate your path to running compliant workloads on Google Cloud.



Cloud Asset Inventory (/asset-inventory/docs)

View, monitor, and analyze Google Cloud and Anthos assets across projects and services.



Organization Policy Service

(/resource-manager/docs/organization-policy/overview)

Centralized and programmatic control over your organization's cloud resources.



Policy Intelligence (/policy-intelligence/docs)

Control resources and manage access through policies to proactively improve your security configuration.



Resource Manager (/resource-manager/docs)

Centralized and programmatic control over your organization's cloud resources.



Risk Protection Program (/resource-manager/docs)

Reduce security risk and gain access to exclusive cyber insurance policies tailored for Google Cloud customers.

Data security



Handle key management for secrets, disks, images, and log retention.



[API keys](/api-keys/docs) (/api-keys/docs)

Use key management for secrets, disks, images, and log retention.



[Cloud External Key Manager](/kms/docs/ekm) (/kms/docs/ekm)

Control the location and distribution of your externally-managed keys.



[Cloud HSM](/kms/docs/hsm) (/kms/docs/hsm)

Protect cryptographic keys with a fully managed hardware security module service.



[Cloud Key Management Service](/kms/docs) (/kms/docs)

Manage encryption keys on Google Cloud.



[Confidential Computing](/confidential-computing/confidential-vm/docs) (/confidential-computing/confidential-vm/docs)

Protect data in-use with Confidential VM, Confidential GKE, Confidential Dataflow, Confidential Managed Service for Apache Spark, and Confidential Space.



[Secret Manager](/secret-manager/docs) (/secret-manager/docs)

Store API keys, passwords, certificates, and other sensitive data.



[Sensitive Data Protection](/sensitive-data-protection/docs) (/sensitive-data-protection/docs)

Discover and redact sensitive data.



[Data Catalog](/data-catalog/docs) (/data-catalog/docs)

Discover and understand your data using a fully managed and scalable data discovery and metadata management service. (Deprecated)

Network security



Centrally manage network resources, establish scalable segmentation for different security zones, and detect network threats.



[Chrome Enterprise Premium](/chrome-enterprise-premium/docs) (/chrome-enterprise-premium/docs)

Use a zero-trust solution that enables secure access with integrated threat and data protection.



[Spectrum Access System](/spectrum-access-system/docs) (/spectrum-access-system/docs)

Manage the wireless communications of devices transmitting in the Citizens Broadband Radio Spectrum (CBRS) band.



[Google Cloud Armor](/armor/docs) (/armor/docs)

Help protect your services against DoS and web attacks.



[Cloud Next Generation Firewall](/firewall/docs) (/firewall/docs)

Implement advanced protection capabilities and pervasive coverage to protect your Google Cloud workloads from internal and external attacks.



[Identity-Aware Proxy \(IAP\)](/iap/docs) (/iap/docs)

Use identity and context to guard access to your applications and VMs.



[Cloud Interconnect](/network-connectivity/docs/interconnect) (/network-connectivity/docs/interconnect)

Connect your infrastructure to Google Cloud on your terms, from anywhere.



[Cloud Intrusion Detection System](/intrusion-detection-system/docs) (/intrusion-detection-system/docs)

Get alerts when Cloud Intrusion Detection System detects malicious activity.



[Cloud VPN](/network-connectivity/docs/vpn) (/network-connectivity/docs/vpn)

Connect your infrastructure to Google Cloud on your terms, from anywhere.



VPC Service Controls (/vpc-service-controls/docs)

Protect sensitive data in Google Cloud services using security perimeters.

Related products, guides, and sites



Gemini Cloud Assist for security and compliance



Understand data protection and privacy.

(/gemini/docs/cloud-assist/analyze-and-optimize-cloud-resources#understand_data_protection_and_privacy_security_controls)

Get help with secrets and Cloud KMS keys.



Get a regulatory control recommendation

(/gemini/docs/cloud-assist/analyze-and-optimize-cloud-resources#get_a_regulatory_control_recommendation)

Get recommendations for which Assured Workload control package to use.

Compliance and privacy



Compliance center (/security/compliance)

View certifications, documentation, and third-party audits to help support your compliance.



Privacy resource center (/privacy)

Discover how we protect the privacy of Google Cloud and Google Workspace customers.



Assured Workloads (/assured-workloads/docs)

Secure your workloads and accelerate your path to running compliant workloads.



Policy violation notifications support

(<https://support.google.com/cloud/answer/7002354>)

View answers to frequently asked questions about Google Cloud policy violations.

[🔗](https://support.google.com/cloud/answer/7002354) (<https://support.google.com/cloud/answer/7002354>)



Data residency service availability (/terms/data-residency)

View a list of services that can be configured for data location. [🔗](#)

(</terms/data-residency>)



Sovereign Controls by Partners

(</sovereign-controls-by-partners/docs>)

Meet digital sovereignty requirements for Google Cloud by Partners.

Secure software supply chain



Software supply chain security

(/software-supply-chain-security/docs/sds/overview)

Use a modular set of Google Cloud products to protect your software supply chain.



Artifact Registry (/artifact-registry/docs)

Store, manage, and secure container images and language packages.



Artifact Analysis (/artifact-analysis/docs/artifact-analysis)

Provide software composition analysis, metadata storage and retrieval.



Assured Open Source Software

(/assured-open-source-software/docs)

Provide enterprise users of open source software with trusted OSS packages.



Cloud Build (/build/docs)

Continuously build, test, and deploy containers using the Google Cloud infrastructure.



[Binary Authorization](/binary-authorization/docs) (/binary-authorization/docs)

Deploy only trusted containers on Google Kubernetes Engine.

Authentication and identity



[Authentication methods](/docs/authentication) (/docs/authentication)

Basics of authentication methods and concepts for Google Cloud services and get help with implementation or troubleshooting.



[Cloud Identity](/identity/docs) (/identity/docs)

Manage user identities, devices, and applications from one console.



[Identity Platform](/identity-platform/docs) (/identity-platform/docs)

Add Google-grade identity and access management to your apps.



[Managed Service for Microsoft Active Directory](/managed-microsoft-ad/docs)

(/managed-microsoft-ad/docs)

Use a highly available, hardened service running Microsoft Active Directory (AD).



Titan Security Keys (/titan-security-key)

Provides phishing-resistant 2nd factor of authentication for high-value users. [↗](#)
(/titan-security-key)



Ensure access and identity.

(https://www.cloudskillsboost.google/course_templates/645)

Study fundamental features of cloud security related to access management and identity. [↗](#) (https://www.cloudskillsboost.google/course_templates/645)

Additional products and resources



Backup and DR Service (/backup-disaster-recovery/docs)

A managed backup and disaster recovery (DR) service for centralized and application-consistent data protection in Google Cloud.



Anti Money Laundering AI

(/financial-services/anti-money-laundering/docs)

Increase anti money laundering detection accuracy and efficiency.



Network Service Tiers (/network-tiers/docs)

Optimize connectivity between systems on the internet and your Google Cloud instances.



Risk Protection Program (/risk-protection-program)

Evaluate your organization's security posture and connect with insurance partners to obtain exclusive cyber insurance coverage and personalized pricing.



Shielded VM (/compute/shielded-vm/docs)

VMs on Google Cloud hardened by a set of security controls that help defend against rootkits and bootkits.

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